This full-text paper was peer-reviewed and accepted to be presented at the IEEE WiSPNET 2017 conference.

A Comparative Study on Adders

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***Abstract*—In a digital circuit, the addition of certain number of bits is generic operation used in order to pare the complexity of the circuit and it operation. The selection of proper adder with requisite properties is more important for the efficient working of the circuit. Comparisons among different adders have been performed, which helps to reduce the laborious work. Adders that have been compared are all of 4 bit and have been synthesized using Xilinx synthesis tool and simulated using the Xilinx simulation tool. The outcomes of synthesis reports and simulation of the circuit helps in finding out the different properties. For paradigm, the area consumed or the number of slices taken up by the circuit and speed can be calculated. These properties make out the difference in the operation and performances of the adders. The adders that have been compared in this paper are Ripple Carry Adder, Carry Look Ahead Adder, Carry Save adder, Carry Skip Adder, Carry select adder, Modified Carry Select Adder and Kogge Stone Adder based on two basic aspects namely, number of slices i.e., area occupied and speed.**

***Index Terms*—Ripple Carry Adder, Carry Look Ahead Adder, Carry Save adder, Carry Skip Adder, Carry select adder, Modified Carry Select Adder and Kogge Stone Adder**

I. Introduction

Adders play crucial role in many arithmetic algorithms of computers. Not only computers, but also used in processors for various operations where the increment of programme counter be one of the example. It has almost become a requisite of most of the cognitive programmes and also considered as pliant. Usage of adders in any circuitry pares the excess number of transistors.

Although adders play a crucial role, based on speed, power dissipation and area consumption, the choice of adder changes from one programme to other [1]. The main requisites of any digital circuits are:

- low power consumption
- low power dissipation
- low area i.e., less number of transistors
- High speed [2].

Since a single adder cannot exhibit all the above mentioned characteristics, certain adders usurp over other based on the requirement of user. In order to make the selection of adders appropriate and not a laborious one, the comparison among various four bit adders has been done extensively.

The adders which have been compared in this paper are:

1. Ripple Carry Adder
2. Carry Look Ahead Adder
3. Carry Save Adder
4. Carry Select Adder
5. Carry Skip Adder

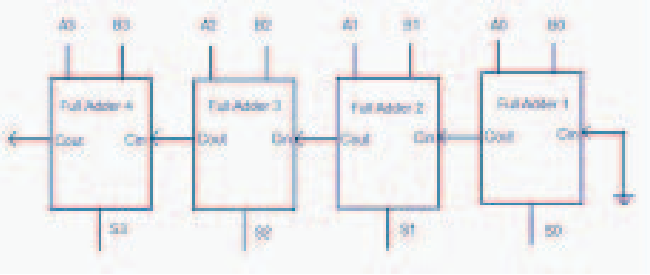


Fig. 1. 4-bit Ripple Carry Adder.

6. Modified Carry Select Adder
7. Kogge Stone Adder.

Section II deals with the functioning including circuit diagrams of respective adders. The Results have been shown and explained in detail in Section III. The comparison of adders regarding various requisites of digital circuits is shown in Section IV. Conclusion of the study done on comparison is specified in Section V.

II. Adders

A. Ripple Carry Adder

Ripple carry adder includes a series of full adders equivalent to the number of bits [3]. The first full adder will be provided with first bits of both two numbers (say A (0) and B (0)) along with input carry (say Cin). The output of first full adder will be the first bit of sum and a carryout, which will be rippled to the next full adder, and so on. Hence the adder called Ripple Carry Adder. Although the area consumption of RCA is less, the delay in the circuit is high. The RCA is the amalgam of low area consumption, high delay time and more power consumption in comparison with other adders that has been mentioned. The Circuit is shown in Fig. 1.

Equations:

$$S = A\ xor\ B\ xor\ Cin$$
$$Cout = AB + BCin + ACin.$$

Looking into the block diagram, the operands that we used are A and B, the operation to be performed is addition and the results are sum(S) and Carry(C). The very first bits of operands i.e., A(0) and B(0) are sent to the first full adder, assuming that the input carry (Cin) is 0. The outputs generated will be the first bit of sum(S(0)) and output carry (Cout) will be rippled to the preceding full adder. Similarly, the second bits of operands are given to the second full adder, third bits are

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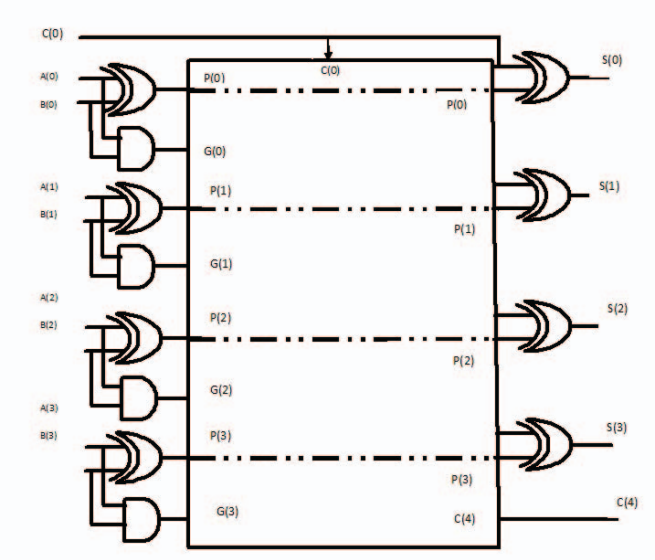


Fig. 2. 4-bit Carry Look Ahead Adder.

given to the third full adder and fourth bits are given to the fourth adder. The sum generated by each full adder will the corresponding bit of sum and the carry generated by each full adder, which will be rippled to the next full adder as input carry. The last full adder provides us the output carry of the operation executed.

B. Carry Look Ahead Adder

The Carry Look Ahead adder provides a better speed in obtaining the result, as the carries in the intermediate stages will be calculated beforehand using carry generate and carry propagate regardless of input carry. Hence named, Carry Look Ahead adder. The extra segments of this adder are the carry propagate and carry generate, where the carry propagate will be propagated to the next stages and the carry generate is responsible for the advance generation of carry irrespective of input carry given to the first stage. The circuit diagram for CLA is shown in Fig. 2.

The propagate and generate terms are obtained based on the equations mentioned above making use of the operands bits i.e., A and B. Later, the carry and sum bits are generated based on the generate and propagate bits developed previously, using the above equations.

Equations of Carry Look Ahead Adder are shown as:

$$P(i) = A(i)\ xor\ B(i)$$
$$G(i) = A(i)\ and\ B(i)$$
$$S(i) = P(i)\ xor\ C(i)$$
$$C(1) = G(0)\ or\ (P(0)\ and\ C(0))$$
$$C(2) = G(1)\ or\ (P(1)\ AND\ G(0))\ or$$
$$(P(1)\ and\ (P(0)\ and\ C(0))$$
$$C(3) = G(2)\ OR\ (P(2)\ and\ G(1))\ or$$
$$(P(2)\ and\ p(1)\ and\ G(0))\ or$$

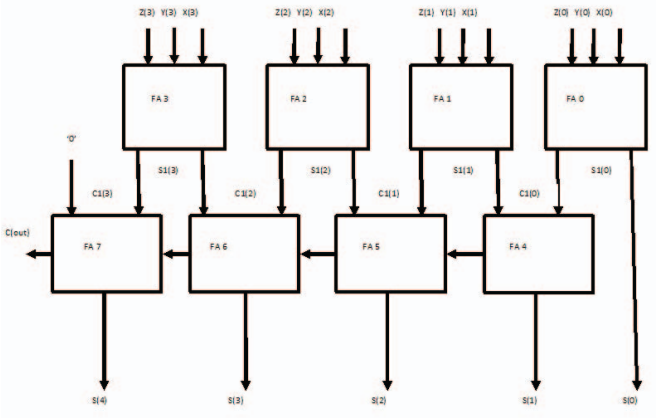


Fig. 3. 4-bit Carry Save Adder.

$$(P(2)\ and\ (P(1)\ and\ P(0))\ and\ C(0))$$
$$C(4) = G(3)\ OR\ (P(3)\ and\ G(2))\ or$$
$$(P(3)\ and\ P(2)\ and\ G(1))\ or$$
$$(P(3)\ and\ P(2)\ and\ P(1))\ and\ G(0))\ or.$$

C. Carry Save Adder

A Carry Save adder is an different one when compared to the previous ones. It doesn't transfer the intermediate carries to the next stages, but instead saves the carry and addends to the sum of next stage using another full adder [5]. This method of adding up bits, generally are 3 binary numbers of 4 bits, pares the time delay of the circuit. The stage1 of the addition part includes saving the carries and sum bits and transferring to the stage 2. The stage 2 acts similar to RCA where the stored carry and sum bits are added. The circuit diagram of the Carry Save adder is as shown in Fig. 3:

The operands used here are three i.e., A, B and Z where Z is a 4 bit input carry. Four full adders are used for four each bits of A, B and Z. The sum and carry bits are generated for every full adder. The carry bits are not sent to the next full adder but instead, they are saved and added up to the next sum term using a ripple carry adder.

D. Carry Skip Adder

Carry Skip Adder uses skip logic in propagation of carry [3]. The basic logic implemented here is that carry is propagated through the bit position unchanged for different values of A1, B1. Here we are taking two RCAs as each block, and the number of RCAs can be extended based on our laborious need. Its logic consists of AND gates which produce carry, and are implemented from second block.

The circuit diagram of Carry Skip Adder is as shown in Fig. 4.

The operands used here are A and B which are of 4 bits. The first two bits of operands are given to the first RCA and the next two bits of operands are given to the second RCA. The carry Cin is initialised as '0' and given to first RCA. The outputs of first RCA are considered as the first two bits of the

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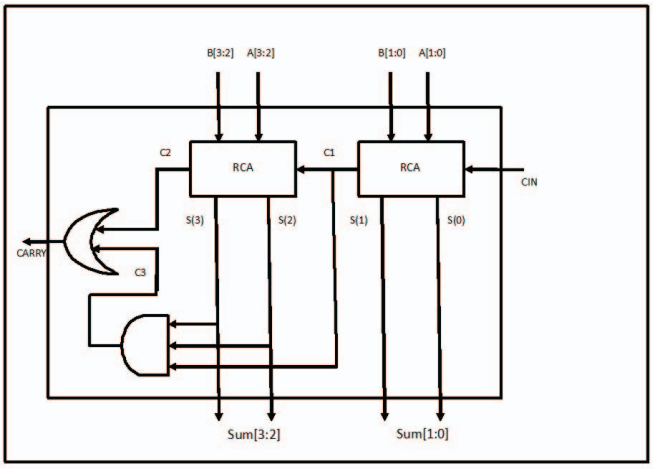


Fig. 4. 4-bit Carry Skip Adder.

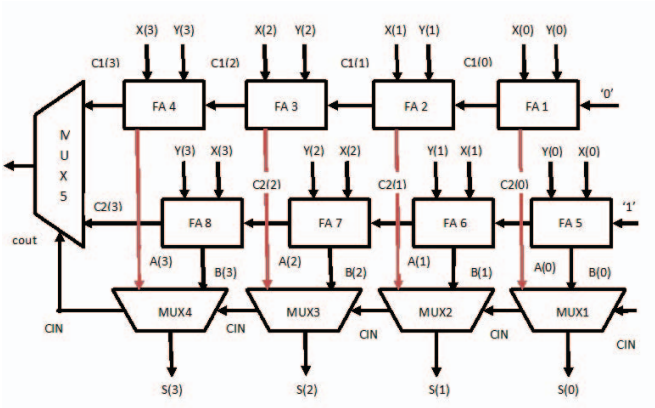


Fig. 5. 4-bit Carry Select Adder.

sum and the output carry obtained is sent to the next RCA, which generates the next two bits of the sum. The overall output carry of the circuit involves AND logic along with an OR gate as shown in the Fig. 4.

E. Carry Select Adder

A Carry select adder generates the output for the possible two values of Cin, i.e. either 0 or 1. It generates the output sum and carry in advance which is done by using two RCAs and the selection of the carry input to be 0 or 1 is done with the help of multiplexer [3]. On a whole, the carry select adder together comprises of two RCAs and a multiplexer.

The circuit diagram of the Carry select adder is as shown in Fig. 5.

The operands, X and Y are given to four full adders along with a input carry equal to 0, and same operands, X and Y are given to other four full adders along with a input carry 1. i.e, we are generating sum and output carry bits with input carry $= 0$ and 1 beforehand. The each group of 4 full adders act as RCA. The output sum bits for input carry $= 0$ and 1 are given to a 2×1 mux where the user provided input carry

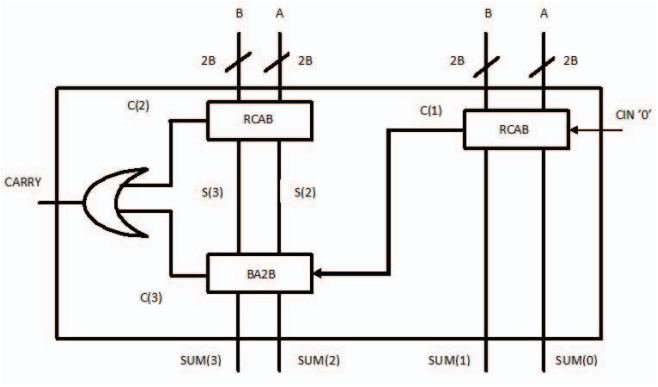


Fig. 6. 4-bit Modified Carry Select Adder.

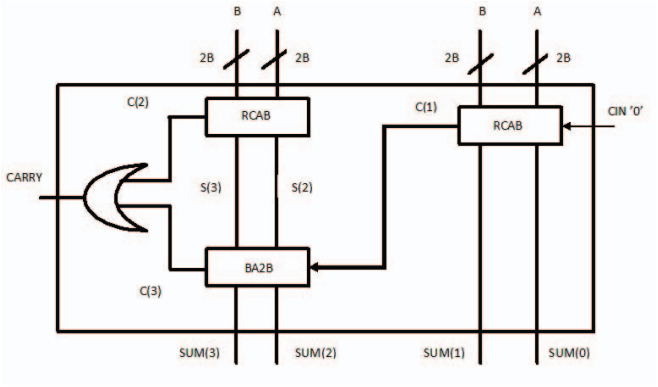


Fig. 7. 4-bit Modified Carry Select Adder.

acts as selection line. The outputs of all the 4 muxs used for sum bits together provide us the total sum. The output carry generated from two RCAs are given to another 2×1 mux where user provided input carry acts as selection line.

F. Modified Carry Select Adder

The Modified Carry Select Adder also called as Conditional BEC-1 consumes less area and operates with high speed when compared with Carry select adder. It makes use of RCA and binary adder which results in simpler circuit. The number of RCAs used in the circuit is half of the total number of bits. The output carry of RCA is added with the sum of the next RCA. The carry out will be obtained by using OR-gate comprising of inputs of last RCA and binary adder.

The circuit diagram of Modified carry select adder is as shown in Figs. 6 and 7.

G. Kogge Stone Adder

The Kogge Stone Adder performs operations with high speed and also considered as parallel prefix form of Carry Look Ahead Adder. This adder completely pares down the time delay in the circuit to generate the carry signals and hence highly used by the circuits or industries that require high speed operations [4]. The circuit diagram of the Kogge Stone Adder is as shown in Fig. 8.

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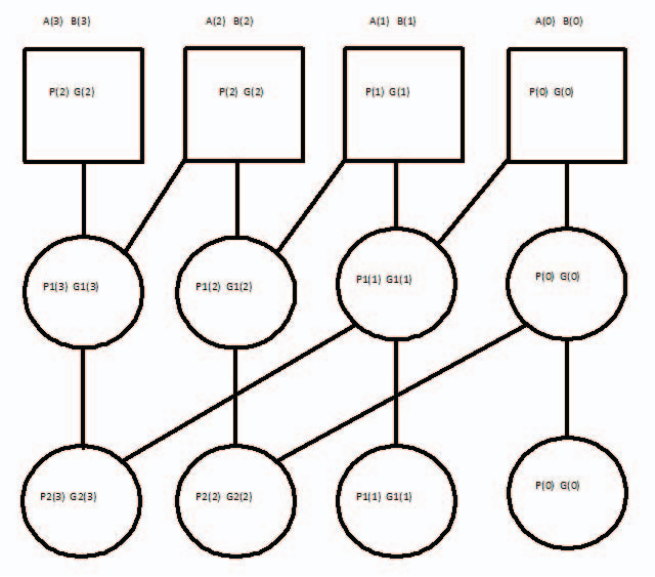


Fig. 8. 4-bit Kogge Stone Adder.

The circuit can be broken down into 3 stages. The stage-1 deals with the generation of P and G using 4 full adders with input carry 0. The stage-2 deals with the generation of P1 and G1 through the above mentioned equations making the use of P and G, where the stage-3 involves the generation of P2 and G2. The propagate and generate terms are obtained for the future prediction of the carries. The sum bits are generated based on the outputs of first stage and generate terms.

III. Results

This section deals with the simulation results of the adders that have been mentioned earlier in Section II. The synthesis process has been done using Xilinx Synthesis tool. The simulation process has been done using the Xilinx simulation tool.

The simulation results give us the information regarding the operands, operation, and its outcomes. The operands here are A and B are of 4 bits and a input carry Cin The values of operands taken for first case are A = "1010", B = "1001", Cin = '0' for first 100 ns which results in the sum as S = "1111" and ouput carry as Co = '0'. The second set of values are A = "1101", B = "0110", Cin = '0' from 100 ns to 200 ns which resulted in the sum bits as S = "0011" and output carry as Co = '1'. The third set of values are A = "1001", B = "1111", Cin = '0' from 200 ns which resulted the sums bits to be S = "1000" and output carry Co = '1'.

The above mentioned values are shown in the form simulation results for various adders as shown in Figs. 9, 10, 11, 12, 13, 14.

The consideration of input carry for carry save adder differs in comparison to others. The output sum bits are of 5 bits and carry out is of 1 bit. The values of operands taken for first case are x = "1010", y = "1001", z = "0000" for first 100 ns to 200 ns which results in the sum as S = "01111" and ouput carry as Co = '0'. The second set of values are X = "1101", Y = "0110", Z = "0000" from 200 ns to 300 ns which resulted in the sum bits as S = "10011" and output carry as Co = '0'. The third set of values are X = "1001", Y = "1111", Z = "0000" from 300 ns to 400 ns which resulted the sums bits to be S = "11000" and output carry Co = '0', as shown in Fig. 15.

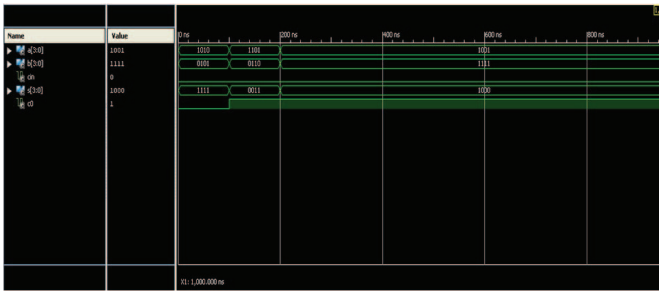


Fig. 9. Simulated waveforms 4-bit RCA.

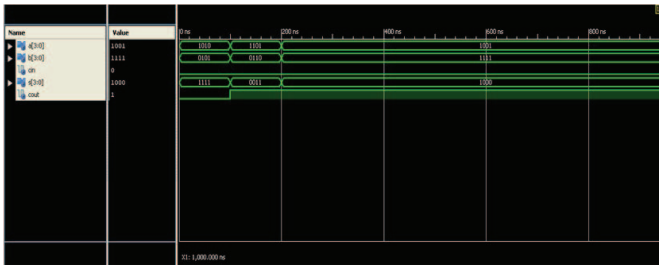


Fig. 10. Simulated waveforms 4-bit CLA.

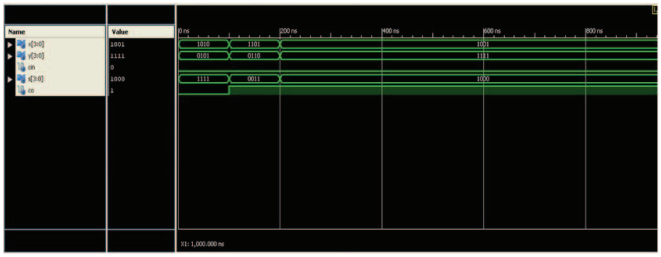


Fig. 11. Simulated waveforms 4-bit Carry Select Adder.

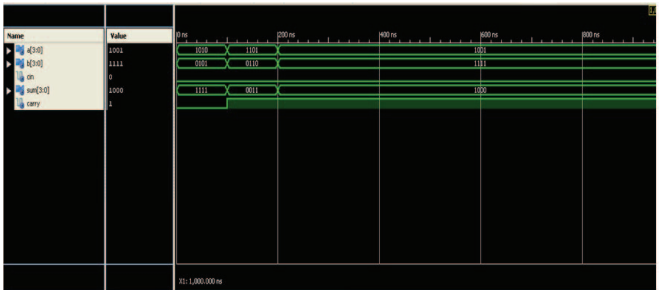


Fig. 12. Simulated waveforms 4-bit Modified Carry Select Adder.

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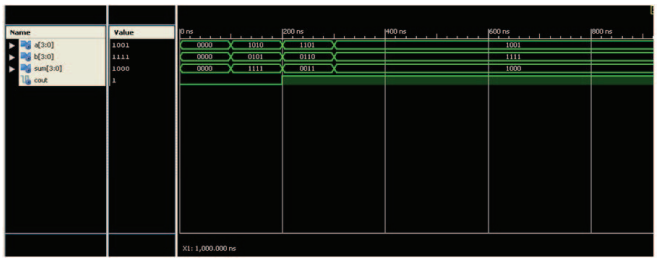


Fig. 13. Simulated waveforms 4-bit Kogge Stone Adder.

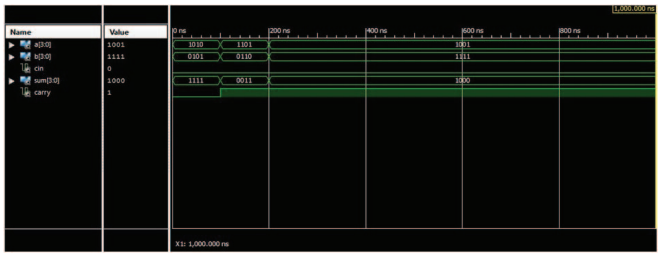


Fig. 14. Simulated waveforms of 4-bit Carry Skip Adder.

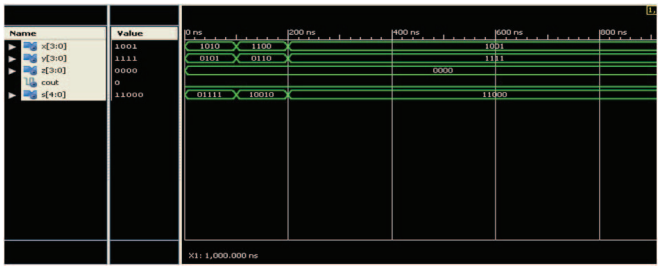


Fig. 15. Simulated waveforms of 4-bit Carry Save Adder.

IV. Comparisons

The comparison table shown below explains the comparison of all the adders mentioned earlier. The comparison is done in three categories i.e., area or number of slices in FPGA, speed of operation. The comparisons have been made under the family of spartan 3E, device named XC3S500E, package of FG320 and speed of -5.

From the comparisons, the ripple carry adder operates with the lowest speed and also occupies less number of slices. Kogge stone adder occupies less number of slices and operates with high speed (see Table I).

TABLE I
Comparison of Adders.

S. No	List of adders	Area/No. of Slices in FPGA (Spartan 3E)	Speed of operation (ns)
1.	Ripple Carry Adder	4	8.959
2.	Carry Look Adder	4	8.920
3.	Carry Save Adder	8	9.196
4.	Carry Skip Adder	6	10.985
5.	Carry Select Adder	5	8.256
6.	Modified Carry Select Adder	5	7.915
7.	Kogge Stone Adder	4	7.820

V. Conclusion

In this comparison study, the comparison is done among all the above mentioned seven 4-bit adders using VHDL codes. As we can observe the differences in the above mentioned comparison table, Ripple Carry Adder consumes less are but takes more time for execution. Carry Look Adder consumes same area as that of RCA but executes the operation in lesser time in comparison. Carry Save Adder consumes more area as well as more time for execution. Carry Skip adder consumes less area in comparison to Carry Save Adder but more when compared to RCA and CLA. Carry select Adder and Modified Carry Select Adder consumes same area, but Modified Carry select adder completes the execution in less time than Carry Select Adder. The Kogge Stone Adder, is the best among all the adders mentioned above, as it requires less area (same as that of RCA) and also requires lesser time than that of RCA. The efficient performance can be obtained using the Kogge Stone Adder.

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